

SOLUTION

- **Variables:** The regions of the map.
- **Domains:** Available colors such as Red, Green, and Blue.
- **Constraints:** Adjacent regions must have different colors.
- Use **backtracking** to assign colors one by one.
- If a selected color violates a constraint, try another color.
- Continue until all regions are colored successfully.

For example, consider regions **A, B, C, D** with the following adjacency:

- A is adjacent to B and C
- B is adjacent to A, C and D
- C is adjacent to A, B and D
- D is adjacent to B and C

Using three colors, a valid assignment can be found.